

MK3

A broadcast channel strip: transformer input, stepped three-band EQ and a fast VCA compressor.



Overview

MK3 is a channel strip in the spirit of the 1970s British local-radio broadcast desks, with a VCA compressor in the spirit of the classic broadcast gain-riding units. The signal runs: input GAIN, phase, input transformer (IRON), high-pass filter, then the EQ and compressor in either order, the output transformer and OUTPUT.

The transformers colour the low end first: the band below about 180 Hz runs into core saturation, so pushing GAIN thickens the bottom before anything else. The EQ has stepped frequencies, like the desk modules, plus FIZZ, a broadcast-style exciter that holds a steady layer of air under the programme.

The compressor is peak-sensing with a fast attack, a small fixed knee and ratios up to infinity. Its release follows the music: spiky material releases faster, dense material slower. Two extras borrowed from broadcast practice - peak reversion correction and the VOICE 2k bell - keep heavy compression from sounding thin or dull.

Reach for it for

- Speech and presenter voice: levelled, clear and forward

- Phone-in and telephone-style tidying
- Drum bus and snare grab with a fast VCA
- Steady bass and broadcast-style piano or acoustic guitar
- Transformer warmth on anything (IRON and GAIN)

Controls

PPM

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A stereo peak programme meter on the output, black face with marks 1 to 7: red needle left, green needle right. 4 is -18 dBFS (alignment) and 6 is -10 dBFS; the marks are 4 dB apart (6 dB between 1 and 2). The needles rise quickly and fall back 24 dB in 2.8 seconds.

INPUT

GAIN

Input gain, -20 to +30 dB. It drives the transformers too, so with IRON in, more GAIN means a thicker, more saturated bottom end.

HPF

A stepped 12 dB per octave high-pass filter: OFF, 40, 63, 100 or 160 Hz. Clears rumble and proximity boom before the EQ and compressor.

Ø

Inverts the polarity of the signal.

IRON

Switches the input and output transformers in or out (in by default). The input transformer saturates the lows first; the output one is lighter.

PEAK

Lights when the signal after GAIN and the input transformer peaks above -3 dBFS.

EQ

Three bands with stepped frequencies. Each band has a small grey FREQ knob and a larger gain knob.

BASS FREQ / BASS	A low shelf at 50, 100 or 200 Hz, -15 to +15 dB.
PRESENCE FREQ / PRES	A bell at 700 Hz, 1, 1.4, 2, 2.8, 4 or 5.6 kHz, -12 to +12 dB.
TREBLE FREQ / TREBLE	A high shelf at 5, 8 or 12 kHz, -15 to +15 dB.
FIZZ	The broadcast 'cut-through' top. An exciter makes new harmonics from everything above 3 kHz, shapes them round 8 kHz and holds them at a steady level about 26 dB under the programme, so the air stays constant and a voice cuts through. It runs in parallel (only the new harmonics are added) and works only when the EQ is IN. The LED shows when it is on.
IN	Switches the EQ in or out (in by default).

COMP

THRESH	The level where compression starts, -40 to 0 dB (-12 dB by default).
RATIO	1:1 to 20:1; fully right reads INF:1, which is limiting.
ATTACK	0.1 to 50 ms (2 ms by default).
RELEASE	20 ms to 3 s (250 ms by default). The release is programme dependent: the compressor watches how spiky the material is and releases faster on spiky material and slower on dense material (from 0.4 to 2.5 times the set value).
GAIN	The compressor's make-up gain, 0 to 24 dB.
VOICE 2k: LIFT / FLAT / DIP	A wide bell at 2 kHz (about two octaves) in front of the compressor's detector, whose depth follows the gain reduction: 0.25 dB for every dB of reduction, up to 4 dB. LIFT raises the voice band as it compresses, so a hard-working compressor doesn't dull the voice; DIP sinks it;

	FLAT (default) leaves it alone. The lamp brightens as the bell comes in.
RELEASE MODE: LOG / LIN	LOG (default): an exponential recovery, fast at first and then easing in. LIN: a steady recovery of 10 dB per release time.
PRC (PEAK REVERSION)	Peak reversion correction, on by default. Under heavy gain reduction it adds back the low end that compression tends to lose, with a gentle shelf around 150 Hz of up to +4 dB that follows the gain reduction.
IN	Switches the compressor in or out (out by default).

Gain reduction, signal flow and output

GR dB	A column of 20 lights showing the compressor's gain reduction, 1 to 20 dB (amber, red from 16 dB).
Signal flow	INPUT, IRON, HPF, EQ, COMP and OUT listed in the order the signal takes; a stage's lamp is lit while it is in use. EQ and COMP swap places with ORDER.
OUTPUT	Output level, -24 to +12 dB.
OUT: OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency, the default), or a lookahead brickwall limiter at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP flashes red while it is clipping.

The header

Preset display	Shows PRESET nn/12 and the preset's name. Click it for the full list. While you move a control, it shows that control's name and value for a moment.
< / >	Step to the previous or next preset.
ORDER	Swaps the EQ and the compressor: EQ > COMP (default) or COMP > EQ. The two units change places on the panel to match.

S / M / L	Panel size: small, medium or large.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
HW	Switches between the hardware-style desk panel and the soft flat panel.
Colour key	The key with the round dot switches the panel between charcoal and cream.

Presets

Init	Every channel control back to its starting value, compressor out.
Presenter Voice	+6 dB gain, HPF 63 Hz, a 2.8 kHz presence lift and a little top; 4:1 compression with a 1 ms attack.
Warm Voice-Over	A touch of 50 Hz weight, 1.4 kHz dipped and 12 kHz air; gentle 3:1 compression.
Interview Guest	HPF 100 Hz, less low end, a 2 kHz lift; firmer 6:1 compression with a fast attack.
Phone-In Tidy	HPF 160 Hz, bass and top cut, 1.4 kHz pushed; heavy 8:1 compression with linear release, PRC off.
Drum Bus Grab	Low and high shelves up, 4:1 with a 5 ms attack and fast linear release.
Snare Smack	200 Hz body and 4 kHz crack; 6:1 with an 8 ms attack to let the hit through, fast linear release.
Bass Even	HPF 40 Hz, 1 kHz dipped; 5:1 compression placed before the EQ (COMP > EQ).
Acoustic Guitar	Transformers out, low end trimmed and 12 kHz sparkle; gentle 2.5:1 compression.

Piano Broadcast	HPF 40 Hz, a little 12 kHz air; light 2:1 compression with a slow release.
Brick Limit	Transformers and EQ out; the compressor at INF:1 with a 0.1 ms attack as a limiter, PRC off.
Iron Warmth	+10 dB into the transformers for low-end thickness, compressor out, OUTPUT at -8 dB to compensate.

Tips

- The presets set the whole channel, including OUTPUT, but leave the OUT stage (OFF / CLIP / LIMIT) as it is.
- For broadcast-style speech, try VOICE 2k on LIFT with PRC on: the voice stays bright and full even when the compressor works hard.
- Put the compressor first (COMP > EQ) when you want the EQ's boosts to stay out of the compressor's way, as in Bass Even.
- Turn FIZZ on for a voice that has to cut through background music or noise; it adds a steady air rather than more level.
- Watch the PPM: keeping the needles around 4 to 6 puts the output around -18 to -10 dBFS.

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